

IOANNIS KALAITZIDIS

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SUMMARY

MSc student in Cybersecurity and Artificial Intelligence Technologies with an academic background in Informatics and Telecommunications. Gained practical exposure to software programming through academic projects, complemented by professional experience in IT operations. This includes 1st/2nd-level IT systems support within the public sector and currently coordinating technical operations, system auditing, and cross-border IT support for an EU-funded initiative (Interreg). Focused on applying theoretical knowledge to practice, demonstrated through multiple hands-on cybersecurity lab projects. Highly motivated to leverage problem-solving skills and technical foundations to protect digital infrastructures in a cybersecurity role.

EDUCATION

MSc, Cybersecurity and Artificial Intelligence Technologies

Expected Graduation March 2027

University of Piraeus

BSc, Informatics and Telecommunications

Graduated July 2025

University of Thessaly

Ranked 1st in Graduating Class

Thesis: "Web-based educational application with a game-based quiz for evaluating and learning Python programming in secondary education" (Grade: 10/10)

Certification: Pedagogical Competence

TECHNICAL SKILLS

Languages: English (C2), Greek (Native)

Programming: Python, C, C++, Java, C#, Bash, HTML, CSS, JavaScript

Cybersecurity and Network Tools: Metasploit, Burp Suite, Wireshark, Nmap, Palo Alto, GDB, Scapy, Ettercap, ModSecurity, aircrack-ng, airmon-ng, Apktool, Ghidra, Frida, Volatility, Autopsy, Mimikatz, Snort

Frameworks, Databases and Dev Tools: Unity, OpenCV, Firebase, Bootstrap, VVenC, scikit-learn, pandas, NumPy

Operating Systems: Windows, MacOS, Linux

Certifications: Introduction to Cloud (Grow with Google - May 2024), Network Basics (CISCO - In Progress), CompTIA Security+ (In Progress)

PROFESSIONAL EXPERIENCE

Technology Expert and Help Desk Coordinator - Rural Research and Development Company of North Evros, Orestiada, Greece

May 2026 – Present

ADAPTAGRI (Interreg NEXT BSB) | Greece-Bulgaria-Türkiye:

- Serve as Technology Expert and Help Desk Coordinator for ADAPTAGRI, an EU co-funded Interreg NEXT Black Sea Basin cross-border programme (Greece-Bulgaria-Türkiye) promoting smart and climate-resilient agriculture across the Evros region.
- Provide 1st and 2nd-level technical support to agricultural end-users adopting precision farming and IoT-based monitoring solutions; coordinate with multi-national project teams to ensure consistent cross-border service delivery.
- Manage technical documentation, user guidance materials and programme deliverables in compliance with Interreg NEXT administrative requirements, supporting smooth project execution under EU reporting standards.

IT Support Trainer and Support - Netcompany, Orestiada-Didymoteicho, Greece

Jan 2026 – May 2026

- Provided on-the-job guidance and comprehensive 1st and 2nd-level IT support to judicial employees in civil court departments and the prosecution office, ensuring a smooth transition to a newly developed digital application.
- Mastered the application's functionality and civil court workflows using official technical manuals, resolving routine user inquiries and actively troubleshooting more complex technical issues.
- Assisted court staff in their day-to-day operations to facilitate smooth software adoption and improve digital efficiency within the public sector.

Military Service: Reserve Second Lieutenant / Research and IT Corps - Hellenic Army, Orestiada, Greece

Nov 2024 – Jan 2026

- Managed information systems reliability and provided comprehensive IT support within the Information Research Office.
- Conducted system monitoring, basic server maintenance and user support to ensure smooth IT infrastructure operations.
- Prepared technical documentation and collaborated with cross-functional teams to troubleshoot technical challenges under strict procedural timelines.

Retail Shop Trainee - COSMOTE, Orestiada, Greece

Jul 2023 – Aug 2023

- Supported daily sales operations and provided high-quality customer service.
- Developed a strong foundational knowledge of modern telecommunications products and technical devices.

ACADEMIC PROJECTS

Cybersecurity Lab Projects and Case Studies (MSc):

Oct 2025 – Present



- **Penetration Testing and Exploitation:** Conducted full-cycle penetration tests based on the Cyber Kill Chain against Windows Server, Heartbleed (CVE-2014-0160) and buffer-overflow targets using Metasploit, Nmap and GDB; performed web-application Red Teaming (SQLi, XSS) via Burp Suite; reverse engineered vulnerable Android applications with Apktool, JADX, Ghidra and Frida to defeat obfuscation and recover hard-coded credentials.
- **Network and Wireless Security:** Executed Man-in-the-Middle attacks (ARP/DNS spoofing, TCP session hijacking) and low-level packet sniffing in Kali Linux using Wireshark, Ettercap and Python (Scapy); completed eight 802.11 case studies spanning monitor-mode frame analysis, deauthentication, WEP/WPS cracking, Evil Twin and rogue-AP deployment, PMKID/KRACK and WPA3 downgrade attacks.
- **Defence-in-Depth and Infrastructure:** Configured Palo Alto firewalls with strict access policies and deployed DNS servers (BIND9) in virtualised environments; implemented layered server defences combining iptables and ModSecurity WAF to block unauthorised access and web attacks.
- **Active Directory, Malware Analysis and Forensics:** Executed end-to-end Active Directory kill chains including Kerberos Golden and Diamond Ticket forgery for persistent domain access; analysed evasive malware attack chains (shellcode injection, Parent PID spoofing) with Sysmon and System Informer; performed disk and memory forensics using Autopsy and Volatility to recover artefacts and reconstruct attacker activity.
- **AI-Powered Security Systems:** Built and evaluated multiple ML-driven security tools - an explainable phishing detector with engineered features (Linear SVC/Random Forest), a hybrid IDS fusing Snort signatures with supervised and unsupervised models ($F1 = 0.926$), a Zero-Trust adaptive access-control engine (Gradient Boosting, Isolation Forest), an LLM-enhanced SSH honeypot with intent classification, and a three-layer prompt-injection defence benchmark reducing attack success from 6.7% to 0% across Llama 3.1 8B and Mistral 7B.
- **Cryptography, Risk Management and Regulatory Analysis:** Analysed the NIST SP 800-232 standard and the Ascon lightweight cryptography family for resource-constrained IoT devices; evaluated ISO/IEC 27005:2022 and NIST risk-management frameworks alongside GRC tooling (SimpleRisk, MONARC, Verinice, EAR/PILAR); co-authored a study on Pegasus and Predator spyware under digital constitutionalism, the EU AI Act and GDPR.
- **Educational Contribution - CyberGame: Hack N Learn:** Participated in an educational cybersecurity initiative by delivering interactive game-based sessions on dumpster diving and SIM card cloning as physical-security and social-engineering attack techniques, raising security awareness among a non-technical audience.

Software Engineering and Web Development Projects (BSc)

Oct 2020 – July 2025

- **Web-Based Educational Platform (Thesis):** Developed a full-stack application with an integrated interactive Unity/C# game to evaluate students' Python skills. Built the front-end using HTML/CSS/JS and Bootstrap, utilizing Firebase for cloud-based user data storage.
- **Game Development:** Created a 2D arcade-style game ("Oh-Mummy" clone) in Python and developed multiple 2D interactive games in Unity utilizing C# scripting.
- **System Programming and Media Processing:** Implemented digital image processing applications in C++ (OpenCV), applied video compression techniques using VVenC and developed Linux system programs utilizing C (pipes, sockets) and Bash scripting.

ACTIVITIES

Volunteer Blood and Bone Marrow Donor

- Registered bone marrow donor (since August 2021) and regular blood donor (since October 2021), demonstrating a continuous and active commitment to public health and social responsibility.